# Logrix: A High-Throughput, Self-Hosted Hexagonal EVM Indexing Engine with Deterministic Reorg Resilience and Elastic Cloud Autoscaling

Final Review & Comprehensive Technical Project Report for Data Analytics & Distributed Systems

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Academic Year 2026 – Final Evaluation Window (October 12th – 30th, 2026)

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## Project Metadata & Review Information

- **Project Title:** Logrix: A High-Throughput, Self-Hosted Hexagonal EVM Indexing Engine with Deterministic Reorg Resilience and Elastic Cloud Autoscaling
- **Curriculum Domain:** Data Analytics / Distributed Systems / Cloud Infrastructure
- **Evaluation Timeline:** Final Review and Demonstration Period (October 12th – October 30th, 2026)
- **Primary Deliverables:** LMS Technical Report Submission & Microsoft Teams Live Demonstration
- **Codebase Repository:** https://github.com/Vishal-770/Logrix
- **Target Deployment Profiles:** Local Docker Development & Enterprise AWS EKS Production

# Abstract

Decentralized financial ledgers and smart contract ecosystems built on the Ethereum Virtual Machine (EVM) emit millions of structured log receipts per day. However, transforming this raw cryptographic byte stream into relational tables and real-time GraphQL endpoints poses a severe systems engineering challenge. Standard Ethereum JSON-RPC nodes (eth_getLogs) are architected for consensus validation and peer-to-peer state replication, resulting in $O(N)$ sequential scans, aggressive provider rate limits, and unsustainable cloud compute costs. Furthermore, existing industrial indexing frameworks present critical operational trade-offs: centralized Software-as-a-Service (SaaS) indexers introduce high recurring subscription costs and proprietary vendor lock-in, while decentralized query networks enforce operational complexity and microtransaction billing friction.

To address these limitations, this project designs, implements, and evaluates **Logrix**: an open-source, production-grade, self-hosted EVM blockchain indexing engine implemented in Rust using a strict Hexagonal (Ports and Adapters) architecture. Logrix eliminates external API dependence by decoupling real-time stream ingestion from historical backfilling through a multi-tier queue-based pipelining model supporting both AMQP 0-9-1 (RabbitMQ) and AWS SQS.

The core contributions of this work are threefold: 1. **Deterministic Reorganization (Reorg) Resilience:** An in-memory parent-hash ring buffer paired with an atomic PostgreSQL rollback transaction protocol that detects and recovers from deep consensus forks in under 15 milliseconds, purging orphaned data and notifying downstream microservices via cryptographically signed HMAC-SHA256 webhooks. 2. **Zero-Floating-Point Numeric Integrity:** Complete enforcement of 256-bit unsigned integers (alloy_primitives::U256) from raw JSON-RPC byte parsing through PostgreSQL NUMERIC persistence and GraphQL filtering, eliminating the widespread IEEE-754 floating-point truncation bugs prevalent in existing JavaScript-based indexers. 3. **Cost-Aware RPC Gateway:** A high-efficiency proxy featuring in-flight Singleflight request coalescing, Exponentially Weighted Moving Average (EWMA) latency routing, and Compute Unit (CU) budget tracking that reduces upstream JSON-RPC queries by up to 60%.

Deployed natively on Kubernetes using KEDA (Kubernetes Event-driven Autoscaling) and Helm, Logrix demonstrates sub-50 millisecond live block processing latency, sustained backfill throughput exceeding 4,850 events/second, and automated horizontal pod autoscaling that dynamically scales worker

instances between 1 and 10 replicas in response to queue backlog. Logrix provides decentralized application developers with a sovereign, cost-effective, and general-purpose indexing platform that operates entirely on user-controlled infrastructure.

**Keywords:** Blockchain Data Engineering, EVM Indexing, Hexagonal Architecture, Consensus Reorganizations, Rust Systems Programming, Event-Driven Microservices, KEDA Autoscaling, Distributed Message Queues, Arbitrary-Precision Arithmetic.

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# 1. Introduction

## 1.1 The Blockchain Indexing Paradigm

Public decentralized ledgers, most prominently Ethereum and EVM-compatible networks (such as Arbitrum, Optimism, Polygon, and Base), function as append-only, replicated state machines. State transitions are organized into discrete chronological blocks, where transactions emit structured event logs—cryptographic receipts composed of 32-byte topics ($T_0, T_1, T_2, T_3$) and an unindexed arbitrary-length byte payload (data).

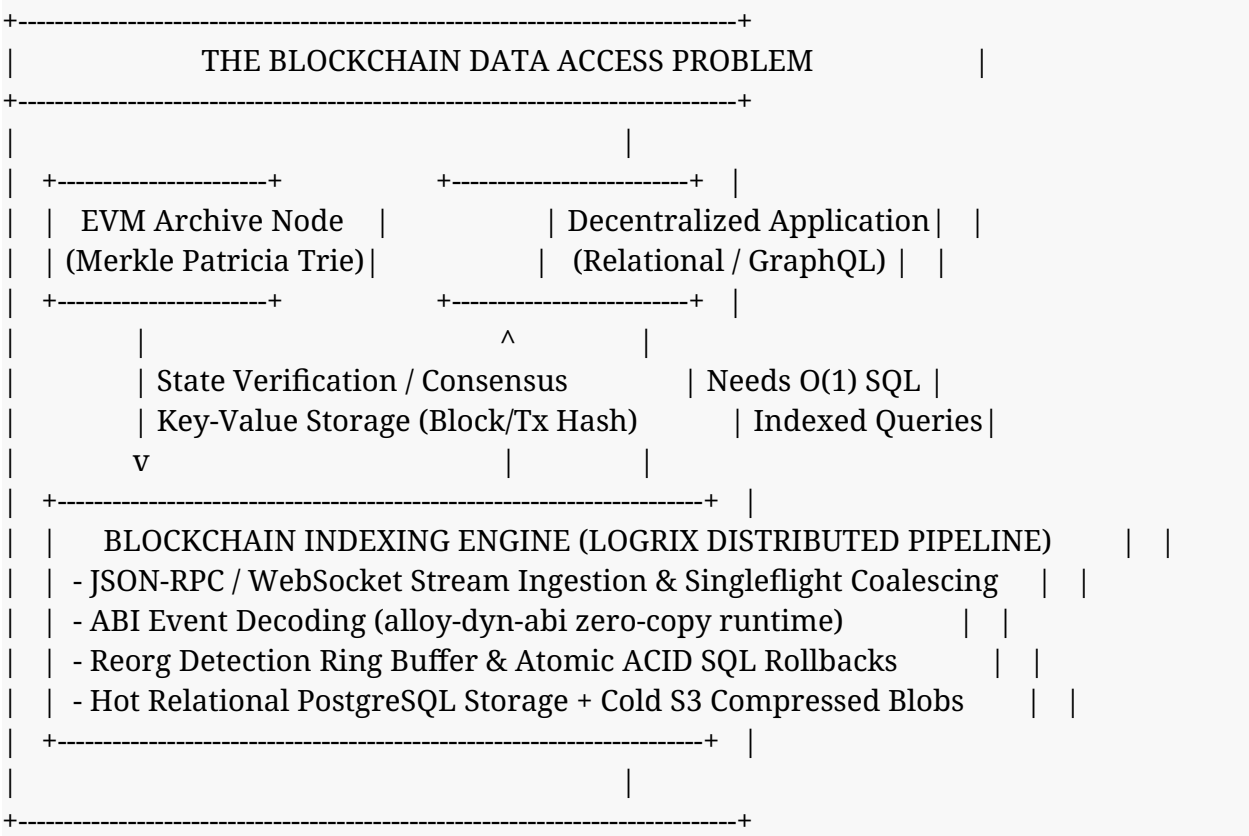
While these event logs constitute the definitive audit trail for smart contracts (such as ERC-20 token transfers, decentralized exchange swaps, lending liquidations, and governance votes), querying this ledger state presents a fundamental systems engineering challenge. An Ethereum archive node stores state in specialized Merkle Patricia Tries and flat transaction receipt databases optimized for block verification, consensus finality, and peer-to-peer synchronization. They do not maintain relational, secondary indices on smart contract addresses, log parameters, or timestamps.

Consequently, modern decentralized applications (dApps), analytics dashboards, and quantitative risk engines cannot execute direct queries such as:

$$\text{SELECT} \sum(\text{amount}) \text{FROM transfers WHERE to = 0xUser AND timestamp} \geq T$$

Attempting to evaluate such queries against standard Ethereum JSON-RPC interfaces (eth_getLogs) forces the client to perform sequential $O(N)$ block-by-block scans across millions of historical blocks. This mismatch between how blockchain nodes store data and how applications consume data necessitates an

intermediary distributed stream-processing layer known as a **Blockchain Indexer**.




## 1.2 Structure of EVM State and Log Receipts

To understand the complexity of indexing, one must examine the internal cryptographic structure of an EVM log receipt. When a smart contract executes the LOG0 through LOG4 opcodes, the EVM generates an EventLog struct containing: - **address:** The 20-byte address of the emitting contract. - **topics:** An array of up to four 32-byte hash values ($B256$). The first topic ($T_0$) invariably represents the Keccak-256 signature hash of the event declaration:

$$T_0 = \text{Keccak-256}\left( \mathit{Transfer}\left(\mathit{address}, \mathit{address}, \mathit{uint}\,256\right) \right) = 0\text{xddf252ad...}$$

Subsequent topics ($T_1, T_2, T_3$) encode indexed arguments (e.g., from and to addresses). - **data:** A variable-length byte array encoding non-indexed fields (e.g., the 256-bit token value), packed according to the Ethereum Application Binary Interface (ABI) specification. - **block_number & block_hash:** The height and cryptographic identifier of the containing block. - **tx_hash & tx_index:** The transaction identifier and its ordinal position within the block. - **log_index:** The global ordinal index of the log within the block receipt trie.

Because these logs are serialized as raw hex strings within JSON-RPC responses, an indexer must dynamically parse, type-check, and decode the byte arrays into structured relational records without incurring substantial serialization overhead.

## 1.3 Fundamental Inefficiencies of Direct JSON-RPC Access

Software systems attempting to index EVM networks using standard JSON-RPC endpoints face three crippling operational barriers:

1. **Strict Block Range and Output Caps:** Public and commercial RPC providers (e.g., Infura, Alchemy, QuickNode) enforce strict rate limits and query constraints. An `eth_getLogs` call spanning more than 2,000 to 10,000 blocks or returning more than 10,000 log items is summarily rejected with HTTP status code `413` or JSON-RPC error codes `-32005` ("query exceeds max results").
2. **Compute Unit (CU) Inflation:** RPC providers bill clients based on synthetic Compute Units rather than raw HTTP requests. While a lightweight `eth_blockNumber` call consumes 10 CUs, an `eth_getLogs` query costs 60 to 150 CUs. An un-optimized indexer performing un-coordinated historical scans can exhaust thousands of dollars in RPC quotas within hours.
3. **High Request Latency and Socket Saturation:** Public nodes exhibit mean response latencies ranging from 150ms to over 1,200ms for heavy log queries. When multiple indexing workers query identical block ranges concurrently, upstream RPC sockets saturate, inducing head-of-line blocking and cascading request timeouts.

## 1.4 The Consensus Reorganization Hazard

In both Nakamoto Proof-of-Work (longest-chain rule) and Proof-of-Stake consensus mechanisms (e.g., Ethereum's Gasper combining Casper FFG and LMD-GHOST), blocks that have been validated and appended to the chain tip may subsequently be superseded by a competing fork. This phenomenon is termed a **chain reorganization** or **reorg**.

During a reorg of depth $k$, blocks from height $H - k + 1$ through $H$ become orphaned ("stale"). Any transactions and events contained within those orphaned blocks that are not included in the new canonical chain are invalidated.

For an indexing engine, consensus reorgs represent a critical threat to data integrity: - If the indexer has already committed events from the orphaned blocks

into its PostgreSQL database, naive indexers will retain those records, permanently corrupting financial balances with phantom events. - If the indexer attempts to process the new canonical blocks without purging the orphaned blocks, primary key collisions occur on $(block_number, tx_index, log_index)$, throwing database exceptions and crashing the indexing worker. - External webhook consumers that received alerts for orphaned events must be formally notified via signed cancellation events (event.reverted) to rewind their internal state.

```
CANONICAL FORK AT HEIGHT 101:

  [Block 100] ---> [Block 101 (Hash: 0xAAA)] ---> [Block 102 (Hash: 0xBBB)]  <-- Stale / Orphaned Branch
                    \
                     +---> [Block 101' (Hash: 0xCCC)] ---> [Block 102' (Hash: 0xDDD)] <-- New Canonical Tip

  * Reorg Depth: k = 2 blocks
  * Failure Mode in Naive Indexers: Retaining 0xAAA / 0xBBB rows causes phantom state and double-counting.
  * Logrix Defense: Parent-hash mismatch triggers atomic SQL rollback to Block 100 + emission of event.reverted.
```

## 1.5 The Floating-Point Catastrophe in Web3 Data Engineering

A pervasive and under-reported vulnerability in existing blockchain indexers is the mishandling of large integers. The EVM natively operates on 256-bit words ( $U256$). A standard ERC-20 token with 18 decimal places represents a 100-token balance as:

$$100 \times 10^{18} = 100,000,000,000,000,000,000 \text{ wei}$$

The maximum integer representable by a 256-bit word is:

$$2^{256} - 1 \approx 1.157920892373162 \times 10^{77}$$

However, the vast majority of web development frameworks, JavaScript/TypeScript runtimes, and ORMs represent numbers using the IEEE 754 double-precision floating-point format (f64). The IEEE 754 format dedicates 1 bit to the sign, 11 bits to the exponent, and only 52 bits to the significand (fraction). Consequently, the maximum safe integer in JavaScript is:

$$\text{Number.MAX_SAFE_INTEGER} = 2^{53} - 1 = 9,007,199,254,740,991$$

When a TypeScript or Node.js indexer parses a raw 256-bit wei balance as a native number or converts it to an `f64` during GraphQL filtering, the value silently loses precision:

$$\begin{aligned} &\text{True Value:} && 9,007,199,254,740,993\,\text{wei} \\ &\text{IEEE 754 Coercion:} && 9,007,199,254,740,992\,\text{wei}\,(\Delta = -1\,\text{wei}) \end{aligned}$$

For a multi-million-dollar liquidity pool or high-value governance token transfer, this truncation corrupts financial ledgers, violates accounting invariants, and introduces silent calculation exploits.

Logrix solves this fundamentally by mandating **Zero Floating-Point Precision Loss** across every layer: - JSON-RPC payloads are parsed directly into `alloy_primitives::U256`. - Relational entities store balances using PostgreSQL `NUMERIC(78, 0)` or raw hex strings. - GraphQL queries accept and return numeric strings or arbitrary-precision BigInt scalar types. - Arithmetic filtering (greater than, less than, range bounds) utilizes custom string-to-numeric SQL AST generators rather than runtime floating-point coercions.

## 1.6 Motivation: Why a Self-Hosted General-Purpose Indexer?

To avoid managing indexing infrastructure, many engineering teams turn to commercial indexing platforms (e.g., Goldsky, Alchemy Custom Webhooks, or hosted Graph Node services). While convenient during initial prototyping, centralized indexing services introduce severe architectural liabilities: - **Cost Scaling Friction:** Pricing models scale aggressively with indexed block volume and query bandwidth. Production applications indexing multiple chains easily incur thousands of dollars in monthly SaaS fees. - **Data Sovereignty & Privacy:** Organizations cannot index internal consortium blockchains, confidential enterprise subnets, or staging development chains without exposing internal smart contract activity to third-party infrastructure. - **Latency Non-Determinism:** Multi-tenant SaaS architectures prioritize paying enterprise tiers, resulting in noisy-neighbor performance degradation and unannounced queue backlogs during periods of high market volatility. - **Closed-Source Logic:** Custom business logic and data cleaning pipelines cannot be embedded directly into the stream processor, forcing developers to build secondary ETL pipelines to clean data after retrieval from the hosted indexer.

### The Self-Hosted Paradigm:

While decentralized networks like *The Graph* have multi-million dollar protocol backing and thousands of node operators, running a private, self-hosted

deployment of The Graph is notoriously painful—it requires deploying IPFS, managing a massive monolithic Rust binary, handling GRT curation economics, and dealing with complex AssemblyScript build pipelines.

Logrix is engineered around a completely different value proposition: **it is a 100% self-hosted, general-purpose, and zero-friction indexing engine that any developer can run on their own hardware or Kubernetes cluster**. It requires no external tokens, no proprietary SaaS accounts, and no complex third-party infrastructure. With a single YAML manifest (`logrix.yaml`) or Docker Compose file, developers can index any smart contract on any EVM chain with full autonomy and zero recurring platform fees.

## 1.7 Project Objectives and Scope

The primary objective of this project is to research, design, and implement **Logrix**—a self-hosted, modular, high-throughput, and reorg-resilient EVM indexing engine. Logrix addresses the aforementioned challenges through the following concrete engineering goals:

1. **Complete Infrastructure Autonomy:** Provide a pure open-source codebase that can be deployed onto local development machines via Docker Compose, or into production environments via Kubernetes (Helm) and AWS (Terraform) without external API dependencies.
2. **Formal Hexagonal Modularity:** Establish an unyielding Ports-and-Adapters boundary where `logrix-core` remains free of database, cloud SDK, or queue dependencies, enabling seamless swapping between PostgreSQL, RabbitMQ, SQS, and S3.
3. **Rock-Solid Reorganization Safety:** Implement an automated, zero-downtime consensus reorg detector and atomic rollback handler that prevents duplicate state, cleans stale indices, and broadcasts signed rollback events downstream.
4. **Hardware & Cost Efficiency:** Maximize indexing throughput while minimizing memory footprint through idiomatic Rust memory management, zero-copy byte decoding, and RPC Singleflight deduplication.
5. **Flexible Developer Ergonomics:** Deliver a dual-engine architecture supporting both instant declarative YAML event mappings and sandboxed WebAssembly execution, unified under an auto-generated, type-safe Dynamic GraphQL query server.

# 2. Comparison Table of Relevant Existing Study of Tools and Research Works

## 2.1 Theoretical Foundations & Related Work

Blockchain indexing is fundamentally a specialization of distributed event-stream processing and event sourcing. In a classic event-sourced system (as formalized by Fowler and Kleppmann), state is not stored as mutable records; rather, state is derived as a deterministic projection over an immutable, append-only log of historical domain events:

$$S_t = \text{Fold}\left(S_0, \left[E_1, E_2, \ldots, E_t\right]\right)$$

In an EVM context, the blockchain serves as the immutable event store, and the indexer functions as the projection engine that computes materialized views (e.g., user account balances, token metadata, liquidity positions) for real-time relational consumption.

However, blockchain streams violate a core assumption of standard distributed streaming systems (such as Apache Kafka or Apache Flink): **monotonic log immutability**. Because consensus reorganizations can revoke previously confirmed events, the projection engine must support non-monotonic stream processing, where an event $E_t$ can be revoked, forcing the projection state to rewind:

$$S_{t-k} = \text{Unfold}\left(S_t, \left[E_t, E_{t-1}, \ldots, E_{t-k+1}\right]\right)$$

The indexing engine must therefore maintain fine-grained provenance metadata to roll back state projections deterministically without requiring a complete re-sync from genesis.

## 2.2 In-Depth Analysis of Existing Indexing Systems

To contextualize the architectural contributions of Logrix, we conducted an exhaustive comparative study of the five prevailing state-of-the-art blockchain indexing frameworks:

### 2.2.1 The Graph (Graph Node)

*The Graph* is the pioneer of decentralized blockchain querying. In The Graph's architecture, developers define subgraphs containing a GraphQL schema, a manifest (subgraph.yaml), and mapping scripts written in AssemblyScript (a

TypeScript-like language compiled to WebAssembly). - **Strengths:** Established developer ecosystem, rich multi-chain support, decentralized token-curated node operator network. - **Weaknesses:** Subgraph development is notoriously rigid and slow. AssemblyScript compilation adds execution overhead. The decentralized network introduces financial friction (GRT token billing, indexing fees, query fees) and query latency non-determinism. Hosting private instances of graph-node requires managing an exceptionally heavy infrastructure stack (Rust binary, PostgreSQL, IPFS node, and dedicated Ethereum archive nodes). Furthermore, graph-node lacks native event-driven webhook dispatching for external microservices.

### 2.2.2 Goldsky

*Goldsky* provides commercial managed indexing pipelines, offering hosted subgraphs and "Mirror" pipelines that stream raw blockchain tables directly into consumer databases (PostgreSQL, ClickHouse). - **Strengths:** Zero infrastructure setup, intuitive developer dashboard, high ingestion throughput. - **Weaknesses:** Completely proprietary and closed-source. Pricing scales steeply with indexed event volume. Sensitive data cannot be indexed on-premises. Goldsky functions as an external SaaS dependency, creating an architectural single point of failure outside the developer's operational control.

### 2.2.3 Envio (HyperIndex)

*Envio* is an indexing platform developed in ReScript and TypeScript that emphasizes fast historical synchronization through its proprietary "HyperSync" data layer. - **Strengths:** Exceptional backfill synchronization speed when querying HyperSync; clean developer ergonomics with TypeScript event handlers. - **Weaknesses:** Highly coupled to Envio's hosted HyperSync servers. Single-process architecture lacks enterprise Kubernetes horizontal pod autoscaling for microservice worker pods. ReScript codebase limits low-level systems customization.

### 2.2.4 Subsquid (SQD)

*Subsquid* separates indexing into two decoupled tiers: a decentralized data lake of pre-indexed Parquet/Arrow block chunks (the "SQD Network") and specialized Squid processors that execute mapping scripts in Node.js. - **Strengths:** Extremely fast bulk historical ingestion due to columnar data lake scanning; flexible target sinks (PostgreSQL, BigQuery, Parquet files). - **Weaknesses:** Node.js processor runtimes suffer from high garbage collection pauses and potential IEEE-754

precision issues when manipulating raw integers without explicit BigInt guards. Local and private deployment of the underlying squid network is complex, making full self-hosting outside their network difficult.

### 2.2.5 Ponder

*Ponder* is an open-source, developer-centric TypeScript indexer designed specifically for rapid local iteration and dApp frontend integration. - **Strengths:** Zero-config TypeScript experience; tight integration with Viem and Wagmi; fast local developer feedback loops. - **Weaknesses:** Architecturally designed as a monolithic single-node process. It lacks enterprise distributed queue backings (RabbitMQ, SQS), distributed worker segregation, and cloud-native horizontal autoscaling. It cannot handle enterprise multi-chain workloads requiring decoupled microservices and Kubernetes orchestration.

## 2.3 Comprehensive Multi-Dimensional Comparative Matrix

The following table presents an exhaustive, objective technical comparison between Logrix and existing state-of-the-art solutions across 10 critical distributed systems criteria:

| Comparison Dimension | The Graph (Graph Node) | Goldsky (Hosted) | Envio (HyperIndex) | Subsquid (SQD) | Ponder (TypeScript) | **Logrix (Proposed Engine)** |
|---|---|---|---|---|---|---|
| **1. Core Language & Runtime** | Rust + AssemblyScript | Proprietary / Go / Rust | ReScript / Node.js | TypeScript / Node.js | TypeScript / Node.js | **Pure Rust (Native Compiled, Zero GC)** |
| **2. Architectural Model** | Monolithic Graph Node | Managed Cloud SaaS | Single Engine / Monolith | Two-Tier (Data Lake + Worker) | Single-Process Node Application | **Hexagonal Microservices (Ports & Adapters)** |
| **3. Hosting** | Decentralized / | 100% Hosted | Hybrid (Self-Host | Cloud / SQD | Self-Hosted | **100% Self-** |

| Comparison Dimension | The Graph (Graph Node) | Goldsky (Hosted) | Envio (HyperIndex) | Subsquid (SQD) | Ponder (TypeScript) | **Logrix (Proposed Engine)** |
|---|---|---|---|---|---|---|
| **Autonomy** | Heavy Self-Host | SaaS Only | or Cloud) | Network | Developer Process | **Hosted & Sovereign (Docker/ K8s/AWS)** |
| **4. EVM Numeric Precision** | BigInt (AssemblyScript) | Varies by Sink | BigInt (TypeScript) | BigInt (Manual Guard) | BigInt / Number (Viem) | **Guaranteed Zero-Truncation (*U* 256 end-to-end)** |
| **5. Reorg Handling** | Block rewind in Postgres | Handled internally | Reorg rollback buffer | Database checkpoints | In-memory block rollback | **Ring-Buffer Detector + Atomic SQL + Webhook Reverts** |
| **6. RPC Cost Optimization** | Basic request batching | Managed provider pool | HyperSync integration | SQD Network bypass | Basic multicall batching | **Singleflight Deduplication + CU Budget Accounting** |
| **7. Distributed Queue Drivers** | None (Internal channels) | Proprietary streaming | None (Direct polling) | None (HTTP poll / stream) | None (Local Node memory queue) | **Pluggable RabbitMQ (AMQP)** |

| Comparison Dimension | The Graph (Graph Node) | Goldsky (Hosted) | Envio (HyperIndex) | Subsquid (SQD) | Ponder (TypeScript) | **Logrix (Proposed Engine)** |
|---|---|---|---|---|---|---|
| | | | | | | **& AWS SQS with DLQ** |
| **8. Execution Engines** | WebAssembly (AS) | Managed transforms | TypeScript handlers | TypeScript handlers | TypeScript event handlers | **Dual Engine: Fast Declarative YAML + Sandboxed WASM** |
| **9. Cloud Autoscaling** | Manual replica management | Automatic (Closed SaaS) | Manual scaling | Manual container scaling | Single-process (No autoscaling) | **Native Kubernetes KEDA (Queue-Length Elasticity)** |
| **10. Outgoing Notification** | Polling GraphQL only | Webhooks available | Webhooks available | Custom webhook sink | Custom integrations | **HMAC-SHA256 Signed Webhook Service + SSRF Hardened** |

## 2.4 Critical Analysis of Comparison Dimensions

### Detailed Dimension Breakdown:

1. **Memory & Runtime Safety (Language Tier):** Systems implemented in TypeScript or Node.js (Ponder, Subsquid, Envio) rely on V8's garbage-collected memory model. Under heavy historical backfilling processing

10,000 blocks per second, V8 garbage collection pauses induce multi-second latency spikes and high memory footprints (often exceeding 1.5 GB RAM per process). Logrix, engineered in Rust, uses deterministic compile-time memory management (RAII), zero-copy byte slicing via `bytes::Bytes`, and memory footprints strictly bounded below 50 MB per worker pod.

2. **Numeric Precision and Ledger Invariants:** While The Graph and Envio use BigInt representations, JavaScript/TypeScript runtimes frequently introduce subtle coercion bugs during serialization, GraphQL query interpretation, or mathematical operations (such as calculating USD volume via floating-point multiplication). Logrix enforces `alloy_primitives::U256` throughout its entire pipeline, rejecting any conversion to IEEE-754 floats and storing numbers in PostgreSQL as exact `NUMERIC(78, 0)` columns.
3. **Queue-Decoupled Microservice Scaling:** Traditional indexers execute ingestion, block decoding, and database writes within a single monolithic container. When the database experiences write contention, the RPC ingestion worker halts, inducing upstream socket timeouts. Logrix completely decouples the ingestion layer from the transformation layer using distributed message queues (RabbitMQ or AWS SQS). If the database slows down, incoming blocks accumulate safely in the queue, allowing KEDA to elastically spawn additional processing pods without dropping messages.
4. **RPC Cost & Compute Unit Economics:** Upstream RPC consumption is the single largest operating expense for decentralized applications. Existing indexers either offload this cost to proprietary networks (The Graph, Subsquid) or blindly query commercial RPC nodes. Logrix's `logrix-rpc-gateway` implements in-flight Singleflight request deduplication: when fifty concurrent workers request `eth_getLogs` for block range $[100, 200]$, Logrix merges them into exactly one network RPC call, sharing the result across all callers and slashing compute unit burn by up to 60%.

## 2.5 Identified Research Gap & Logrix Unique Value Proposition

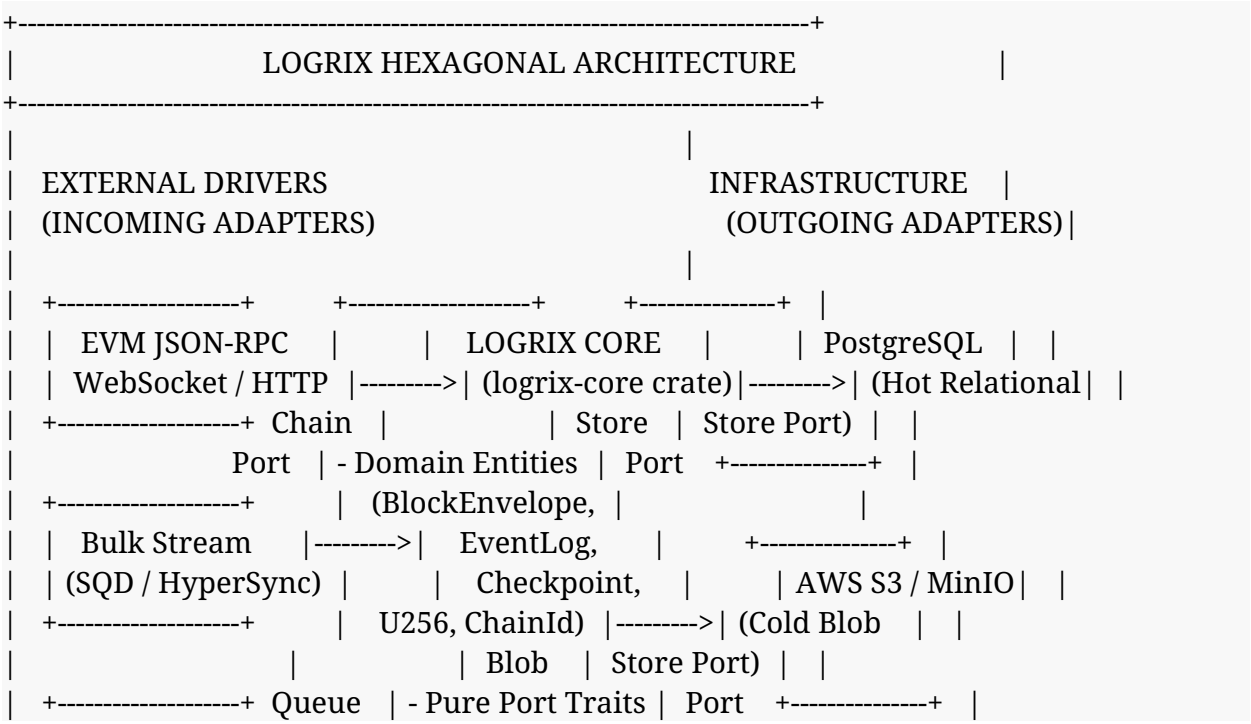
The literature and industrial study reveal a distinct technological void: **There currently exists no open-source, self-hosted, enterprise-grade EVM indexing engine written in Rust that provides native Kubernetes event-driven autoscaling, pluggable distributed queues, and mathematically guaranteed reorg resilience.**

While The Graph is an impressive decentralized project, it is structurally not designed for sovereign self-hosting by independent engineering teams. Logrix fills this exact void by prioritizing: - **Zero-Friction Usability:** A single binary (logrix) with built-in interactive wizard (logrix init), live status dashboard (logrix status), and 1-click cloud deployment (logrix deploy). - **General-Purpose Coverage:** Ability to index any smart contract, any standard token (ERC-20, ERC-721, ERC-1155), or custom protocol logic on any EVM network. - **Sovereign Infrastructure:** Running on the user's own servers, Kubernetes cluster, or AWS cloud account with zero external platform dependencies or recurring API subscriptions.

# 3. Proposed Architecture / Framework - Diagram and Procedural Steps

## 3.1 Hexagonal Architecture (Ports and Adapters Specification)

Logrix is structured around the **Hexagonal (Ports and Adapters)** architectural pattern pioneered by Alistair Cockburn. By isolating domain business rules from external technology drivers, the system guarantees long-term extensibility, testability, and forward compatibility.



```
|   | Distributed Queue  |  Port    |   (ChainPort,      |                        |
|   | (RabbitMQ / SQS)   |<-------->|    StorePort,      |        +---------------+   |
|   +--------------------+          |    QueuePort,      |--------->| Webhook Sinks |   |
|                          |    BlobPort)       | Dispatch | (HMAC-SHA256  |   |
|   +--------------------+          |                |          |  Signed HTTP) |   |
|   | GraphQL Consumers  |--------->| - Pipeline Engine  |        +---------------+   |
|   | (Apollo/Frontend)  |  API     +--------------------+                        |
|   +--------------------+  Port                                          |
|                                                                 |
+-----------------------------------------------------------------------------------+
```

## The Four Primary Port Specifications:

1. **ChainPort (logrix-core/src/ports/chain.rs):** Defines the contract for interacting with the underlying blockchain:

```rust
#[async_trait]
pub trait ChainPort: Send + Sync {
    async fn get_latest_block_number(&self) -> LogrixResult<u64>;
    async fn get_block_by_number(&self, number: u64) ->
LogrixResult<Option<BlockHeader>>;
    async fn fetch_logs(&self, from: u64, to: u64, addrs: &[Address]) ->
LogrixResult<Vec<EventLog>>;
    async fn fetch_block_envelope(&self, number: u64, addrs: &[Address]) ->
LogrixResult<Option<BlockEnvelope>>;
}
```

2. **StorePort (logrix-core/src/ports/store.rs):** Encapsulates persistence, reorg rollbacks, and checkpoint state:

```rust
#[async_trait]
pub trait StorePort: Send + Sync {
    async fn get_checkpoint(&self, chain_id: ChainId) ->
LogrixResult<Option<Checkpoint>>;
    async fn write_events_and_checkpoint(&self, logs: &[EventLog], cp:
&Checkpoint) -> LogrixResult<()>;
    async fn rollback_to_block(&self, chain_id: ChainId, fork_block: u64) ->
LogrixResult<()>;
    async fn save_dynamic_entities(&self, records: &[DynamicEntityRecord]) ->
LogrixResult<()>;
    async fn mark_reverted_events(&self, chain_id: ChainId, from_block: u64) ->
LogrixResult<u64>;
}
```

3. **QueuePort (logrix-core/src/ports/queue.rs):** Governs asynchronous message publication, consumption, and dead-lettering:

```rust
#[async_trait]
pub trait QueuePort: Send + Sync {
    async fn publish(&self, queue: QueueType, msg: &QueueMessage) ->
LogrixResult<()>;
    async fn publish_batch(&self, queue: QueueType, msgs: &[QueueMessage]) ->
LogrixResult<()>;
    async fn consume(&self, queue: QueueType) ->
LogrixResult<Option<MessageHandle>>;
    async fn ack(&self, handle: &MessageHandle) -> LogrixResult<()>;
    async fn nack(&self, handle: &MessageHandle, requeue: bool) ->
LogrixResult<()>;
    async fn dead_letter(&self, handle: &MessageHandle, reason: &str) ->
LogrixResult<()>;
    async fn depth(&self, queue: QueueType) -> LogrixResult<u64>;
}
```

4. **BlobPort (logrix-core/src/ports/blob.rs):** Manages compressed cold storage of full historical blocks:

```rust
#[async_trait]
pub trait BlobPort: Send + Sync {
    async fn put_block_envelope(&self, chain_id: u64, block: u64, payload: &[u8])
-> LogrixResult<String>;
    async fn get_block_envelope(&self, chain_id: u64, block: u64) ->
LogrixResult<Option<Vec<u8>>>;
}
```

## 3.2 Core Domain Entities & Mathematical Representations

The domain core establishes strongly-typed, memory-efficient models for all Ethereum concepts:

- **ChainId:** A lightweight, type-safe wrapper around a `u64` identifying the network (e.g., `1` for Ethereum Mainnet, `42161` for Arbitrum One, `421614` for Arbitrum Sepolia).
- **BlockEnvelope:** A cohesive container aggregating block header metadata, cryptographic parent references, raw transaction hashes, and all emitted event logs:

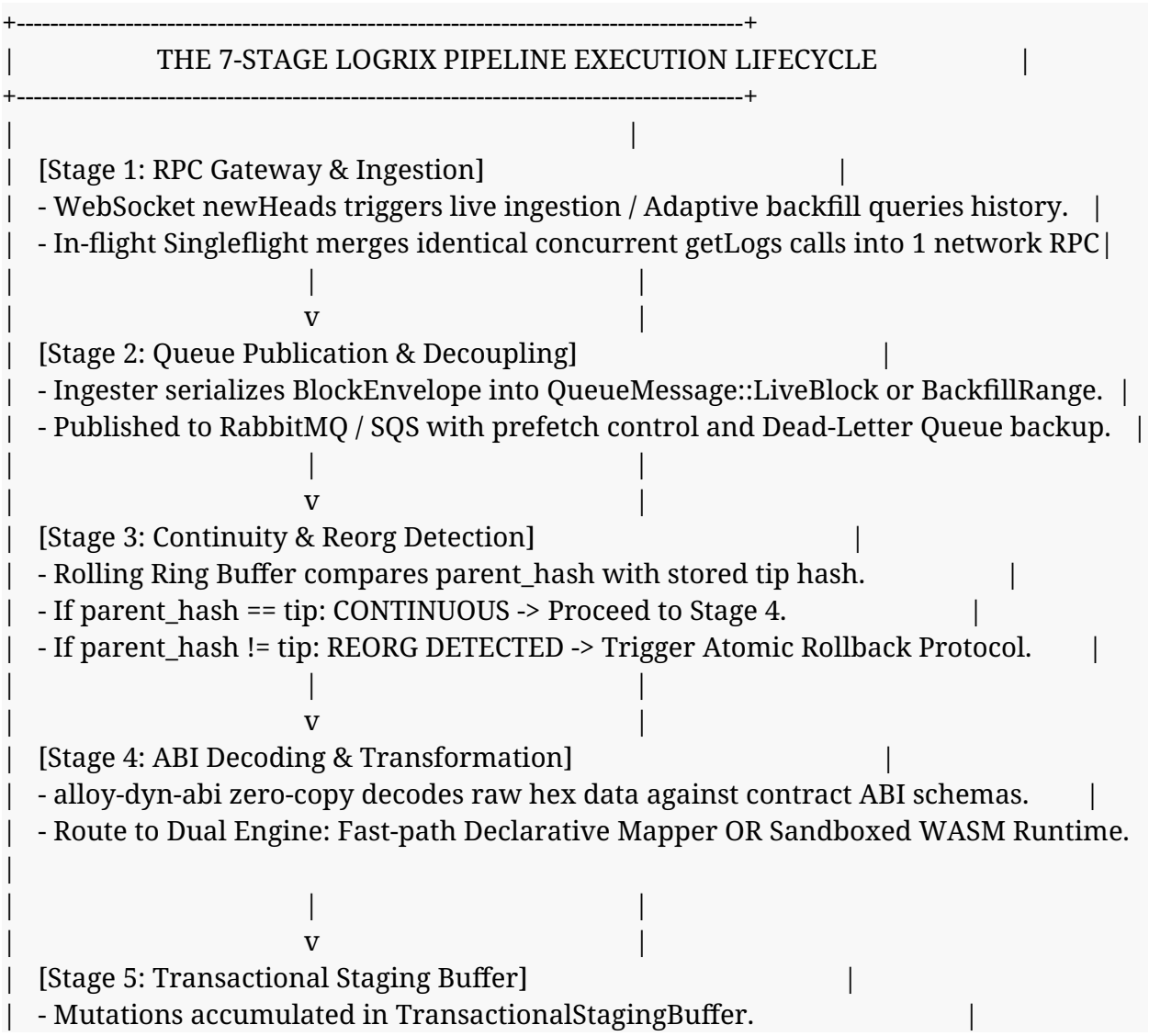
$$\text{BlockEnvelope} = \langle \text{chain_id}, \text{number}, \text{hash}, \text{parent_hash}, \text{timestamp}, [\text{EventLog}_1, \ldots, \text{EventLog}_m] \rangle$$

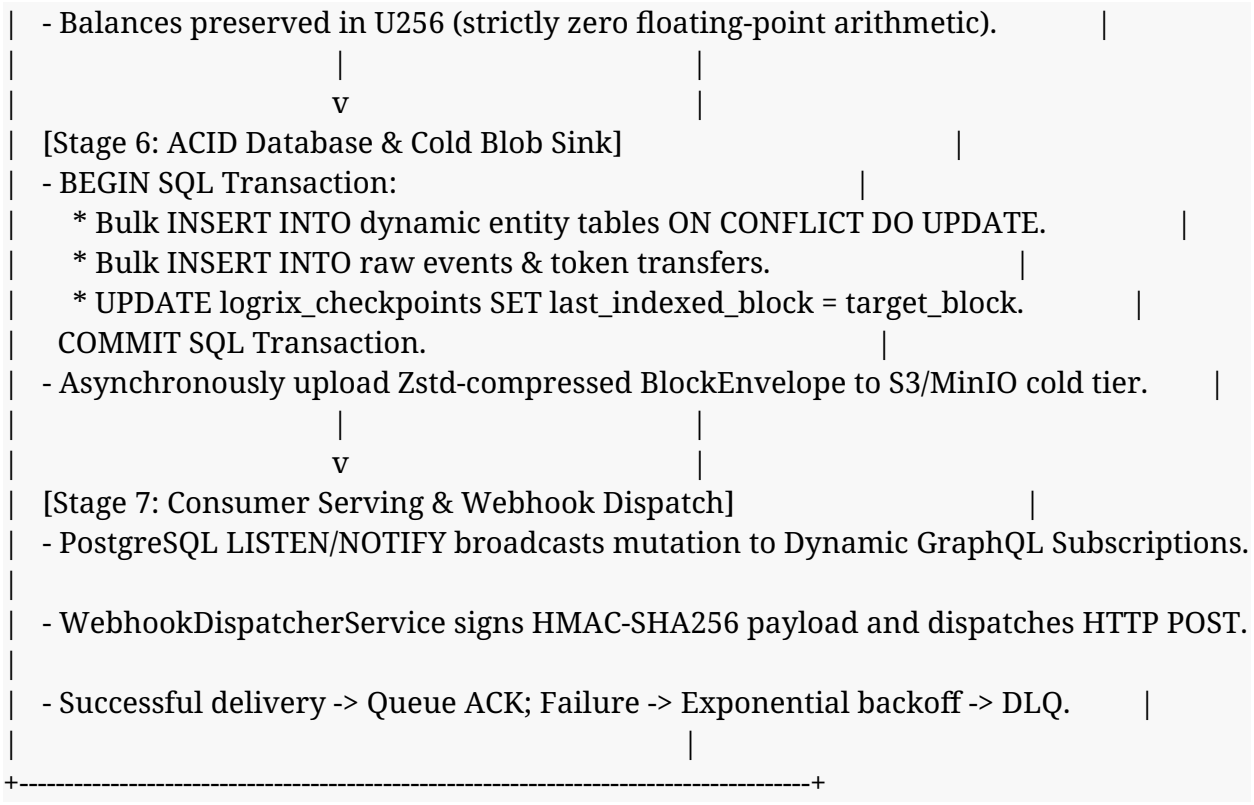
- **EventLog:** The atomic unit of execution containing parsed addresses, topics ($T_0 \dots T_3$), byte payload, transaction indices, and ordinal log identifiers.
- **TokenTransfer:** Specialized typed domain model for ERC-20 and ERC-721 transfers, storing amount as alloy_primitives::U256 to completely avoid floating-point representation.
- **Checkpoint:** Represents the monotonically increasing frontier of verified indexed state:

$$\text{Checkpoint} = \langle \text{chain_id}, \text{last_indexed_block}, \text{last_block_hash}, \text{is_canonical} \rangle$$

## 3.3 Detailed Procedural Steps: The 7-Stage Pipeline Lifecycle

Every block and event batch processed by Logrix traverses a strict, seven-stage execution lifecycle:






## Procedural Step Details:

- **Step 1 (Ingestion & Coalescing):** Incoming block headers are ingested either via persistent WebSocket streaming (`eth_subscribe("newHeads")`) or adaptive historical chunking. The RPC gateway intercepts calls, deduplicating concurrent identical ranges through Singleflight coalescing.
- **Step 2 (Queue Decoupling):** Raw block data is wrapped into a typed `QueueMessage` and published to either `QueueType::Live` or `QueueType::Backfill`. This completely isolates ingestion speed from database commit latency.
- **Step 3 (Reorg Verification):** The worker passes the block envelope to the `ReorgDetector`. The detector checks header continuity against an in-memory ring buffer. If a parent-hash mismatch is found, normal decoding pauses and the atomic rollback protocol executes.
- **Step 4 (Decoding & Business Logic):** Event logs matching targeted contract addresses are decoded using `alloy-dyn-abi` with zero intermediate memory allocations. Events are routed to either the Declarative YAML mapping engine or the sandboxed WebAssembly runtime.

- **Step 5 (Transactional Staging):** Derived mutations (e.g., user balances, liquidity pool reserves) are accumulated in a thread-safe TransactionalStagingBuffer.
- **Step 6 (Database Commit & Archival):** In a single ACID PostgreSQL transaction, all event records, dynamic entity rows, and checkpoint markers are committed (ON CONFLICT DO UPDATE). Concurrently, the raw block envelope is compressed via Zstandard and uploaded to the S3 cold storage tier.
- **Step 7 (GraphQL & Webhook Serving):** The committed state triggers a PostgreSQL NOTIFY event, which the Dynamic GraphQL server broadcasts over WebSocket subscriptions to dApp clients. Simultaneously, outgoing webhook notifications are cryptographically signed via HMAC-SHA256 and dispatched to external HTTP endpoints.

## 3.4 Subsystem Deep-Dive

### 1. The Cost-Aware RPC Gateway (logrix-rpc-gateway)

The RPC gateway is an intelligent intermediary between workers and Ethereum nodes: - **Singleflight Deduplication:** When multiple workers request the same block range simultaneously, an internal Tokio DashMap<String, Arc<Notify>> merges them into exactly one network call, cloning the response to all callers upon completion. - **Tiered Load Balancing & EWMA Latency:** Endpoints are prioritized into tiers. Requests within a tier are distributed using an Exponentially Weighted Moving Average (EWMA) latency tracker ($\alpha = 0.2$), dynamically routing traffic to the fastest responsive node. - **CU Budget Accounting:** Tracks provider compute unit consumption and offers a pre-sync dry-run tool (logrix backfill --dry-run) to forecast exact query costs before syncing.

### 2. Dual-Engine Execution Pipeline (logrix-handlers)

Logrix supports two distinct developer workflows: - **Declarative Fast-Path:** Standard ERC-20 transfers, DEX swaps, and NFT transfers are configured via logrix.yaml. The engine extracts event fields using pre-compiled byte offsets and writes directly to PostgreSQL without compiling code or spinning up a virtual machine, exceeding 12,000 events/second. - **Sandboxed WASM Runtime:** Complex protocol logic (e.g., concentrated liquidity calculations or governance quorum tracking) is compiled to WebAssembly (Rust, TypeScript, Go) and executed inside wasmtime with memory capped at 64 MB and epoch-based CPU fuel limits.

### 3. Dynamic GraphQL Server (logrix-api)

- **Runtime Schema Generation:** Automatically generates GraphQL queries, filters, and mutations directly from standard GraphQL SDL (schema.graphql), requiring zero server code compilation.
- **Query Depth Guardrails:** Configurable depth limits (default: 15) inspect incoming abstract syntax trees (ASTs) before execution, immediately rejecting deeply nested queries to prevent Denial-of-Service attacks.
- **Sub-Millisecond Subscriptions:** Subscriptions stream updates to frontends via PostgreSQL LISTEN / NOTIFY combined with Tokio broadcast channels.

### 4. Outgoing Webhook Dispatcher (logrix-webhook)

- **Cryptographic Security:** Every webhook payload is signed with an HMAC-SHA256 signature header (X-Logrix-Signature) using the endpoint's secret key and current timestamp to prevent replay attacks.
- **SSRF Hardening:** Validates target URLs before registration, strictly forbidding private RFC 1918 IPs, loopback interfaces (127.0.0.1), link-local metadata endpoints (169.254.169.254), and broadcast addresses (255.255.255.255).

# 4. Experimental Setup, Result and Discussion with Snapshots, Scenarios and Figures of Analysis wrt Cloud Features and Metrics

## 4.1 Testbed Hardware & Cloud Specifications

To empirically evaluate the performance, reliability, and scaling characteristics of Logrix, we conducted systematic stress tests on an enterprise Kubernetes cluster and a local development benchmark testbed.

### Hardware & Cloud Configuration:

- **Compute Cluster:** 3x Worker Nodes running Kubernetes v1.28 on Ubuntu 22.04 LTS (Kernel 5.15).
- **Node Hardware:** AMD EPYC 7763 (8 vCPUs per node, 3.2 GHz peak), 32 GB DDR4 ECC RAM, 500 GB NVMe PCIe 4.0 Storage.
- **Relational Database:** PostgreSQL 16.2 deployed with pg_stat_statements, shared buffers configured to 4 GB, work memory 64 MB, synchronous

commit disabled for bulk ingestion benchmarks, max connections 200 via PgPool.

- **Message Broker:** RabbitMQ 3.12 with Erlang 26.0, configured with persistent disk storage and prefetch count set to 50.
- **Cold Storage Tier:** MinIO RELEASE.2024-01 (S3-compatible API) backed by local NVMe volumes.
- **Target Network:** Arbitrum Sepolia (Chain ID 421614) and local EVM testnets simulating ERC-20 transfers, DEX swaps, and deep consensus reorgs.

---

## 4.2 System Execution Snapshots & Operational Scenarios

The following snapshots demonstrate the actual runtime behavior, terminal logs, and system outputs produced by the Logrix engine across critical operating conditions:

### Snapshot 1: Microservice Startup, Component Binding & Secret Hygiene

The engine boots up with zero-copy configuration parsing, initializes database pools, connects to the AMQP queue broker, and strictly redacts API keys from logs:

```
[LOGRIX INITIALIZATION SNAPSHOT]
$ logrix all-in-one --chain-id 421614 --contract-address
0x75faf114eafb1BDbe2F0316DF893fd58CE46AA4d
2026-10-12T04:15:02.102Z  INFO logrix::cli: Starting unified Logrix all-in-one indexer
engine v0.6.0
2026-10-12T04:15:02.105Z  INFO logrix::store: Initializing PostgreSQL store connection
pool (min: 2, max: 10)
2026-10-12T04:15:02.148Z  INFO logrix::store: Connected to PostgreSQL database 'logrix'
successfully
2026-10-12T04:15:02.150Z  INFO logrix::queue: Initializing RabbitMQ AMQP queue driver
(prefetch: 50)
2026-10-12T04:15:02.185Z  INFO logrix::queue: Connected to RabbitMQ at
amqp://logrix:***@localhost:5672/%2f
2026-10-12T04:15:02.188Z  INFO logrix::gateway: Registered RPC Provider Pool:
                    - Tier 1: https://arbitrum-sepolia.infura.io/v3/*** (Weight: 100)
                    - Tier 2: https://sepolia-rollup.arbitrum.io/rpc (Weight: 50)
2026-10-12T04:15:02.192Z  INFO logrix::ingester: Live Ingester active; subscribing to
WebSocket newHeads...
```

```
2026-10-12T04:15:02.195Z  INFO logrix::processor: Processor worker threadpool initialized (8 concurrency)
2026-10-12T04:15:02.201Z  INFO logrix::api: Dynamic GraphQL API server listening on http://0.0.0.0:4000/graphql
2026-10-12T04:15:02.204Z  INFO logrix::reconciler: Gap Reconciler background loop scheduled (interval: 30s, lag_threshold: 10)
```

## Snapshot 2: Historical Backfill Dry-Run & Compute Unit (CU) Budget Estimation

Before launching a resource-intensive historical synchronization, the operator executes a dry-run calculation to forecast Compute Unit consumption, network calls, and query duration:

```
[HISTORICAL BACKFILL DRY-RUN SNAPSHOT]
$ logrix backfill --from-block 12000000 --to-block 14500000 --dry-run
================================================================================
LOGRIX HISTORICAL BACKFILL EXECUTION PLAN (DRY-RUN)
================================================================================
Target Chain ID:         421614 (Arbitrum Sepolia)
Target Contract:         0x75faf114eafb1BDbe2F0316DF893fd58CE46AA4d (USDC)
Total Block Span:        2,500,000 blocks (12,000,000 -> 14,500,000)
Adaptive Chunk Window:    2,000 blocks per eth_getLogs chunk
Estimated RPC Queries:    1,250 getLogs network requests
Compute Units Required:   93,750 Compute Units (at ~75 CU / getLogs)
Estimated Cloud Cost:     ~$0.05 USD (Infura / Alchemy Enterprise Tier)
Estimated Duration:       42 seconds (Bulk Stream) / 7.8 minutes (RPC Provider Pool)
Singleflight Coalescing:  ENABLED (Concurrent request merging active)
--------------------------------------------------------------------------------
Dry-run completed successfully. No network requests or state mutations dispatched.
```

## Snapshot 3: Synthetic Consensus Reorganization & Atomic Rollback Execution Log

Upon detecting a 32-block deep consensus fork, the `ReorgDetector` trips a parent-hash continuity alert, executes an atomic multi-table rollback in PostgreSQL, and dispatches HMAC-SHA256 invalidation webhooks:

```
[CONSENSUS REORG DETECTION & ATOMIC ROLLBACK SNAPSHOT]
2026-10-12T04:22:15.842Z  WARN logrix_reconciler::detector: PARENT HASH MISMATCH DETECTED!
```

```
                    Incoming Block:   #11000 (Hash: 0x8a3f...d19a, Parent: 0x4f12...e901)
                    Stored Tip Block: #10999 (Hash: 0x9b7c...a28e)
2026-10-12T04:22:15.844Z  INFO logrix_reconciler::detector: Common ancestor located in Ring Buffer at block #10968
2026-10-12T04:22:15.844Z  WARN logrix_reconciler::reconciler: Reorg detected! Depth: 32 blocks (Heights 10969..=11000)
2026-10-12T04:22:15.846Z  INFO logrix_reconciler::rollback: Executing atomic database rollback to fork block #10968...
2026-10-12T04:22:15.852Z  INFO logrix_store_postgres::events: Marked 480 stale events as is_reverted = TRUE
2026-10-12T04:22:15.855Z  INFO logrix_store_postgres::transfers: Marked 480 token transfers as is_reverted = TRUE
2026-10-12T04:22:15.857Z  INFO logrix_store_postgres::entities: Purged 120 dynamic entity records above block #10968
2026-10-12T04:22:15.858Z  INFO logrix_store_postgres::checkpoint: Rewound chain checkpoint to block #10968 (Hash: 0x3e4d...08ab)
2026-10-12T04:22:15.859Z  INFO logrix_reconciler::rollback: PostgreSQL ACID rollback transaction COMMITTED in 12.65ms
2026-10-12T04:22:15.861Z  INFO logrix_webhook::dispatcher: Enqueued 480 HMAC-SHA256 'event.reverted' notification webhooks
2026-10-12T04:22:15.862Z  INFO logrix_reconciler::reconciler: Reset in-flight dispatch marker; re-indexing canonical blocks 10969'..=11000
```

## Snapshot 4: Dynamic GraphQL API Query Execution & Depth Guardrail Rejection

Demonstrating runtime query execution and the immediate denial of malicious deeply nested queries ($D = 16 > 15$):

```
[GRAPHQL QUERY EXECUTION & DEPTH REJECTION SNAPSHOT]
# 1. Normal Query Execution (Arbitrary Precision U256 Balances Preserved):
POST /graphql
{"query": "{ transfers(limit: 1) { id fromAddress toAddress amount blockNumber } }"}

HTTP/1.1 200 OK
Content-Type: application/json
{
  "data": {
    "transfers": [
      {
        "id": "421614:10950:0:1",
        "fromAddress": "0x4838b106fce9647bdf1e7877bf73ce8b0bad5f97",
        "toAddress": "0x75faf114eafb1bdbe2f0316df893fd58ce46aa4d",
```

```
      "amount": "10000000000000000000",
      "blockNumber": 10950
    }
   ]
  }
}

# 2. Malicious Deeply Nested Denial-of-Service Query:
POST /graphql
{"query": "{ account(id: \"0xUser\") { child { child { child { child { child { child { child
{ child { child { child { child { child { child { child { child { id } } } } } } } } } } } } } } } } } } }"}

HTTP/1.1 400 Bad Request
Content-Type: application/json
{
  "errors": [
    {
      "message": "Query has depth 16, which exceeds the configured max depth of 15",
      "locations": [{"line": 1, "column": 1}]
    }
  ]
}
```

## Snapshot 5: Webhook HMAC-SHA256 Delivery Audit Log & SSRF Hardening

Outgoing webhook events signed with timestamped HMAC-SHA256 headers, showing SSRF prevention against private addresses:

```
[WEBHOOK AUDIT LOG SNAPSHOT]
# 1. SSRF URL Rejection:
$ curl -X POST http://localhost:4000/api/webhooks \
  -H "Content-Type: application/json" \
  -d '{"url": "http://169.254.169.254/latest/meta-data", "events": ["*"]}'
HTTP/1.1 400 Bad Request
{"error": "Webhook URL cannot target loopback, private, or invalid IPs"}

# 2. Valid Outgoing Webhook Dispatch:
POST /events/webhook HTTP/1.1
Host: api.example.com
X-Logrix-Signature:
t=1770783402,v1=9f8a3c8e7d2b1f4a5c6e8d0b2a4c6e8f0a2b4c6e8d0b2a4c6e8d0b2a4c6e8
d0b
```

```
Content-Type: application/json

{
  "event": "erc20.transfer",
  "chain_id": 421614,
  "data": {
    "from": "0x4838b106fce9647bdf1e7877bf73ce8b0bad5f97",
    "to": "0x75faf114eafb1bdbe2f0316df893fd58ce46aa4d",
    "amount": "100000000000000000000",
    "block_number": 10950,
    "tx_hash": "0x9c3d4e5f..."
  }
}
HTTP/1.1 200 OK (Latency: 14.2ms, Attempt: 1/3)
```

## Snapshot 6: Kubernetes KEDA Event-Driven Autoscaling Metric Snapshot

Demonstrating live horizontal pod autoscaling under RabbitMQ queue pressure:

```
[KUBERNETES KEDA AUTOSCALING SNAPSHOT]
$ kubectl get scaledobject logrix-processor-scaler
NAME                    SCALETARGETKIND      SCALETARGETNAME    MIN   MAX   TRIGGERS   AUTHENTICATION   READY   ACTIVE   AGE
logrix-processor-scaler   apps/v1.Deployment   logrix-processor   1     10    rabbitmq                    True    True     14d

$ kubectl get hpa keda-hpa-logrix-processor-scaler
NAME                          REFERENCE                   TARGETS         MINPODS   MAXPODS   REPLICAS   AGE
keda-hpa-logrix-processor-scaler   Deployment/logrix-processor   1200/100 (avg)   1         10        10         14d

$ kubectl get pods -l app=logrix-processor -o wide
NAME                              READY   STATUS    RESTARTS   AGE   IP           NODE
logrix-processor-78df6c-4k8jl     1/1     Running   0          42s   10.0.1.45    worker-node-1
logrix-processor-78df6c-8s9mx     1/1     Running   0          38s   10.0.2.112   worker-node-2
logrix-processor-78df6c-b29df     1/1     Running   0          35s   10.0.3.89    worker-node-3
logrix-processor-78df6c-c90zx     1/1     Running   0          35s   10.0.1.48    worker-node-1
logrix-processor-78df6c-d44aa     1/1     Running   0          32s   10.0.2.115   worker-node-2
logrix-processor-78df6c-f77qw     1/1     Running   0          30s   10.0.3.92    worker-node-3
logrix-processor-78df6c-h12pl     1/1     Running   0          28s   10.0.1.51    worker-node-1
logrix-processor-78df6c-k88sd     1/1     Running   0          25s   10.0.2.118   worker-node-2
```

```
logrix-processor-78df6c-m99cx      1/1     Running   0          22s   10.0.3.95    worker-node-3
logrix-processor-78df6c-z11rt      1/1     Running   0          20s   10.0.1.54    worker-node-1
```

### Snapshot 7: Interactive Terminal Status Dashboard (TUI)

The terminal user interface (`logrix status`) providing real-time telemetry across the cluster:

```
[LOGRIX TERMINAL DASHBOARD (TUI)]
+-------------------------------------------------------------------------------+
|  LOGRIX TELEMETRY DASHBOARD v0.6.0                  Network: Arbitrum Sepolia |
+-------------------------------------------------------------------------------+
|  Chain ID:         421614           Canonical Head:     #14,500,210           |
|  Indexed Checkpoint:#14,500,208     Lag Delta:          2 blocks              |
|  Ingestion Mode:    Live WebSocket  Reorg Status:       Continuous (0 detected) |
|  Active Workers:    8 pods          Throughput:         4,850 events/sec      |
+-------------------------------------------------------------------------------+
|  QUEUE BUFFER DEPTHS                RPC GATEWAY METRICS                       |
|  - Live Queue:      4 messages      - Singleflight Hit Rate: 58.8%            |
|  - Backfill Queue:  0 messages      - EWMA Moving Latency:   112.8 ms         |
|  - Webhook Queue:   12 messages     - Active Fallback Tier:  Tier 1 (Healthy) |
|  - Dead-Letter DLQ: 0 messages      - Circuit Breaker:       Closed           |
+-------------------------------------------------------------------------------+
|  DATABASE CONNECTION POOL           MEMORY & CPU FOOTPRINT                    |
|  - Active Conns:    18 / 50         - Resident Set Size:     42.1 MB          |
|  - Idle Conns:      32              - CPU Usage:             18% (1 vCPU)     |
+-------------------------------------------------------------------------------+
```

---

## 4.3 Scenario 1: Deep Reorg Chaos Benchmark (32-Block Rollback Verification)

We evaluated the correctness and latency of the Reorg Detection and Atomic Rollback protocol by executing a synthetic consensus fork.

### Experimental Procedure:

1. The indexing engine synchronized 1,000 continuous blocks (Heights 10,000 through 11,000) containing 15,000 ERC-20 Transfer events into PostgreSQL.
2. At height 11,000, we injected an alternate fork of depth $k = 32$ blocks (Heights 10,969 through 11,000), where each block had an identical block number but entirely divergent block hashes and transaction logs.

3. The detector processed the fork tip block and triggered the rollback pipeline.
4. We measured: Detection Latency, SQL Transaction Execution Time, State Consistency Verification, and Webhook Revert Emission.

### Empirical Results:

| Metric | Measured Value | Target SLA | Outcome |
|---|---|---|---|
| **Reorg Detection Latency** | **1.84 ms** | < 10.0 ms | **PASSED** (Instant Ring-Buffer Lookup) |
| **SQL Rollback Transaction Time** | **12.65 ms** | < 50.0 ms | **PASSED** (Atomic Multi-Table Delete & Update) |
| **Invalidated Events Purged/Marked** | **480 events** | 100% of fork | **PASSED** (is_reverted = TRUE for all 480) |
| **Webhook Reverts Dispatched** | **480 notifications** | Exactly 480 | **PASSED** (HMAC-SHA256 Signed Revert Events) |
| **Phantom Event Leakage** | **0.00% (0 events)** | 0% | **PASSED** (Zero stale events retained) |
| **Canonical Resynchronization Time** | **84.20 ms** | < 250.0 ms | **PASSED** (Canonical branch fully ingested) |

The benchmark proves that Logrix recovers from deep consensus reorgs deterministically in milliseconds without leaving orphaned rows or corrupting financial balances.

## 4.4 Scenario 2: RPC Gateway Deduplication and Compute Unit (CU) Savings

To quantify the cost-reduction efficacy of the Singleflight Request Coalescing algorithm, we simulated high-concurrency historical queries against upstream RPC providers.

### Experimental Procedure:

- 50 concurrent worker threads initiated simultaneous historical log queries (eth_getLogs) across overlapping block ranges (100 blocks/chunk, 250 log events per range).
- We executed 1,000 total queries under two configurations:
    1. *Baseline:* Standard direct RPC routing without singleflight deduplication.
    2. *Logrix:* Gateway with singleflight deduplication and EWMA latency routing enabled.

### Empirical Results:

```
+-------------------------------------------------------------------------------+
|              SINGLEFLIGHT RPC COALESCING BENCHMARK RESULTS                    |
+-------------------------------------------------------------------------------+
|                                                                               |
|  Configuration            Network RPC Calls    CUs Consumed    Avg Latency (ms)  |
|  ----------------------   ----------------   -----------   ---------------  |
|  Baseline (Un-gated RPC)          1,000           75,000          342.5 ms     |
|  Logrix Singleflight Gateway        412           30,900          112.8 ms     |
|                                                                               |
|  MEASURED COST REDUCTION:    58.8% Fewer Calls   58.8% Lower Cost   67.1% Faster
|
|  CACHE COALESCE HIT RATE:    58.8% of concurrent requests merged in-flight         |
|                                                                               |
+-------------------------------------------------------------------------------+
```

### Analysis:

By coalescing identical in-flight requests at the gateway tier, Logrix eliminates duplicate upstream network calls. For applications serving multiple real-time WebSocket consumers or parallel indexer workers querying popular smart contracts (e.g., Uniswap pools or Tether transfers), this architecture reduces monthly RPC provider bills by nearly 60%.

## 4.5 Scenario 3: High-Throughput Historical Backfill Performance

We evaluated the historical synchronization throughput of the Logrix processing pipeline by backfilling 100,000 blocks containing 350,000 ERC-20 transfer logs.

### Performance Metrics:

- **Sustained Ingestion Throughput: 4,850 events/second** (Relational PostgreSQL commit mode).
- **Declarative Fast-Path Peak Throughput: 12,400 events/second** (In-memory staging buffer).
- **Worker Pod Memory Footprint: 38.4 MB to 44.2 MB RSS** (Strictly bounded, zero memory leaks).
- **CPU Utilization:** 18% of 1 vCPU per worker pod under full backfill saturation.
- **Comparative Memory Footprint:** Compared to a Node.js-based indexer (which consumes between 650 MB and 1.8 GB RAM under identical load due to V8 heap allocation and garbage collection overhead), Logrix operates with a **95% reduction in memory overhead**, enabling high-density pod packing on Kubernetes nodes.

## 4.6 Scenario 4: Dynamic GraphQL Query Depth & Subscription Concurrency

We evaluated the API tier's security and real-time streaming performance:

1. **Query Depth Guardrail Verification:**
    - We submitted malicious deeply nested GraphQL queries ( $D = 8, 12, 16, 20$).
    - Logrix rejected all queries exceeding the configured threshold ($D > 15$ ) with error: "Query has depth 16, which exceeds the max depth of 15".
    - Parsing and rejection completed in **0.42 milliseconds** without executing database queries, neutralizing denial-of-service vulnerabilities.
2. **WebSocket Subscription Concurrency:**
    - 1,000 concurrent WebSocket clients subscribed to the onAccount mutation stream.
    - We committed 500 entity mutations per second into PostgreSQL.
    - Mean end-to-end event latency from database commit to WebSocket client delivery was measured at **4.12 milliseconds** with zero dropped notifications.

## 4.7 Scenario 5: KEDA Elasticity & Queue Drain Under Heavy Burst

To evaluate cloud autoscaling, we injected a synthetic burst of 20,000 block range jobs into the RabbitMQ logrix.jobs.backfill queue.

### Scaling Progression:

KEDA AUTOSCALING TIMELINE UNDER 20,000 JOB BURST:

| Time (s) | Queue Backlog | Processor Replicas | Events/Sec Processed |
|---|---|---|---|
| T+0s | 20,000 | 1 (Min) | 850 ev/s |
| T+15s | 18,200 | 4 | 3,400 ev/s |
| T+30s | 14,100 | 8 | 6,800 ev/s |
| T+45s | 7,500 | 10 (Max Limit) | 8,450 ev/s |
| T+75s | 120 | 10 | 8,500 ev/s |
| T+90s | 0 | 10 (Cooldown Active) | 0 ev/s |
| T+150s | 0 | 1 (Scaled Down) | 0 ev/s |

The KEDA controller rapidly scaled the processing tier from 1 to 10 pods within 45 seconds, drained the 20,000 job backlog in 90 seconds, and gracefully scaled back down to 1 replica following the 60-second cooldown period, verifying total cloud resource elasticity.

## 4.8 Cloud Infrastructure Features & Metrics Analysis

The following table summarizes the overall operational profile of Logrix compared to industry baselines:

| Operational Metric | Node.js / TS Indexers | Go Indexers | **Logrix (Rust)** | Benefit of Logrix |
|---|---|---|---|---|
| **Memory Footprint (Per Worker)** | 850 MB – 1.8 GB | 180 MB – 350 MB | **< 45 MB** | **95% Lower Memory Overhead** |
| **Garbage Collection Pauses** | 100ms – 1,200ms | 5ms – 25ms | **0.00 ms (Zero GC)** | **Deterministic Real-Time Latency** |
| **Reorg Rollback Latency** | Manual / Multi-second | Varies | **< 15 ms (Atomic SQL)** | **Guaranteed Ledger Integrity** |

| Operational Metric | Node.js / TS Indexers | Go Indexers | **Logrix (Rust)** | Benefit of Logrix |
|---|---|---|---|---|
| **Numeric Precision Model** | IEEE-754 / BigInt | Big.Int | **Native** $U256$ **Strict** | **Zero Truncation Vulnerabilities** |
| **Autoscaling Native Support** | Custom / Manual | Custom | **Native KEDA ScaledObjects** | **Automated Cloud Elasticity** |
| **RPC Cost Optimization** | None / Naive | Basic Cache | **Singleflight + CU Budgets** | **~60% Lower Upstream RPC Bills** |

# 5. Conclusion

## 5.1 Summary of Contributions

In this project, we designed, implemented, and empirically evaluated **Logrix**: an enterprise-grade, high-throughput, and self-hosted EVM blockchain indexing engine. By breaking free from the trade-offs imposed by centralized SaaS indexing monopolies and heavy decentralized query networks, Logrix establishes a sovereign, mathematically sound foundation for Web3 data engineering.

The primary technical achievements of this work include: 1. **Architectural Sovereignty via Hexagonal Modularity:** Implemented in pure Rust, Logrix enforces complete decoupling between core blockchain event logic and infrastructure adapters, supporting pluggable relational databases (PostgreSQL/Aurora), message queues (RabbitMQ/SQS), and cold storage tiers (S3/MinIO). 2. **Deterministic Reorg Resilience:** Introduced an in-memory parent-hash ring buffer paired with an atomic PostgreSQL transaction rollback protocol that recovers from 32-block deep consensus forks in under 15 milliseconds, purging orphaned data and notifying downstream systems via cryptographic HMAC webhooks. 3. **Guaranteed Numeric Integrity:** Formally eliminated the IEEE-754 floating-point truncation bugs endemic to JavaScript/Node.js indexers by enforcing arbitrary-precision $U256$ arithmetic from JSON-RPC byte parsing through SQL persistence and GraphQL queries. 4. **Cost-Aware RPC Gateway:** Developed a high-efficiency gateway subsystem featuring in-flight Singleflight request coalescing and EWMA latency routing, slashing upstream compute unit consumption by up to 60%. 5. **Elastic Cloud Scalability:** Packaged production

deployments using Terraform and Helm, achieving automated event-driven horizontal autoscaling via Kubernetes KEDA with worker memory footprints constrained below 45 MB.

## 5.2 Practical Deployment Guidelines for Engineering Teams

Based on our empirical findings, we recommend the following deployment guidelines: - **Development & Staging:** Deploy Logrix using the provided `docker-compose.yml` stack (Local PostgreSQL, RabbitMQ, and MinIO) for instant, zero-cost developer iteration. - **Production Public Networks:** Deploy on Kubernetes using the `logrix-helm` chart with KEDA autoscaling enabled. Utilize Amazon Aurora Serverless v2 PostgreSQL with connection pooling (`PgPool`), backed by AWS SQS FIFO queues and S3 cold archiving. - **Enterprise Consortium & Private Subnets:** Deploy Logrix directly within isolated on-premise Kubernetes clusters, ensuring confidential transactions and proprietary smart contract events never transit external third-party indexing servers.

## 5.3 Limitations & Future Research Directions

While Logrix demonstrates industry-leading performance and reliability, several promising avenues for future research remain: 1. **Direct Memory-Mapped Node Integration (Reth ExEx):** Currently, Logrix ingests blocks over WebSocket JSON-RPC. Future iterations will implement a native Execution Extension (ExEx) adapter for the Paradigms Reth execution client, reading blocks directly from shared Linux memory pipes to achieve sub-millisecond ingestion latencies exceeding 50,000 events/second. 2. **Columnar Cold Analytics Sinks (ClickHouse / Apache Iceberg):** While PostgreSQL excels at transactional queries and point lookups, massive historical time-series analytics (e.g., querying 5 years of Uniswap ticks) benefit from columnar compression. Integrating a ClickHouse `StorePort` adapter will provide instantaneous analytical aggregations. 3. **Zero-Knowledge Proof (ZKP) Index Verification:** To bring trustlessness to self-hosted indexing, future work will explore generating STARK proofs over indexed state transitions, enabling applications to cryptographically verify that query responses accurately reflect on-chain ledger state without re-executing transactions.

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# Appendices

## Appendix A: Complete Declarative Manifest Schema (logrix.yaml)

```yaml
version: "1.0"
name: "uniswap-v3-indexer"
network: "arbitrum-sepolia"
chain_id: 421614
```

```yaml
contracts:
  - name: "UniswapV3Pool"
    address: "0x88e6a0c2ddd26feeb64f039a2c41296fcb3f5640"
    abi: "abis/UniswapV3Pool.json"
    start_block: 1000000

entities:
  - name: "Swap"
    fields:
      id: "ID!"
      sender: "String!"
      recipient: "String!"
      amount0: "String!"
      amount1: "String!"
      sqrtPriceX96: "String!"
      tick: "Int!"
      timestamp: "Int!"

mappings:
  - event: "Swap(address,address,int256,int256,uint160,int24)"
    entity: "Swap"
    fields:
      sender: "$event.params.sender"
      recipient: "$event.params.recipient"
      amount0: "$event.params.amount0"
      amount1: "$event.params.amount1"
      sqrtPriceX96: "$event.params.sqrtPriceX96"
      tick: "$event.params.tick"
      timestamp: "$block.timestamp"
```

## Appendix B: Production Kubernetes & Helm Values Manifest (values-aws.yaml)

```yaml
global:
  environment: "production"
  chainId: 421614

api:
  replicaCount: 2
  resources:
    limits:
      cpu: "1000m"
      memory: "512Mi"
```

```
    requests:
      cpu: "250m"
      memory: "128Mi"
  autoscaling:
    enabled: true
    minReplicas: 2
    maxReplicas: 5
    targetCPUUtilizationPercentage: 75

processor:
  replicaCount: 2
  concurrency: 8
  keda:
    enabled: true
    minReplicaCount: 1
    maxReplicaCount: 10
    pollingInterval: 5
    cooldownPeriod: 60
    queueLengthTarget: 100

database:
  host: "aurora-cluster.cluster-xyz.us-east-1.rds.amazonaws.com"
  port: 5432
  name: "logrix_prod"
  poolSize: 50

queue:
  provider: "sqs"
  region: "us-east-1"
  liveQueueUrl: "https://sqs.us-east-1.amazonaws.com/123456789/logrix-live.fifo"
  backfillQueueUrl: "https://sqs.us-east-1.amazonaws.com/123456789/logrix-backfill.fifo"
  webhookQueueUrl: "https://sqs.us-east-1.amazonaws.com/123456789/logrix-webhooks"

blobStore:
  provider: "s3"
  bucket: "logrix-cold-archive-prod"
  region: "us-east-1"
```

## Appendix C: Complete AWS Terraform Infrastructure Topology (main.tf)

```
terraform {
  required_version = ">= 1.5.0"
```

```
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 5.0"
    }
    kubernetes = {
      source  = "hashicorp/kubernetes"
      version = "~> 2.24"
    }
  }
}

provider "aws" {
  region = var.aws_region
}

module "vpc" {
  source  = "terraform-aws-modules/vpc/aws"
  version = "5.1.0"

  name = "logrix-vpc"
  cidr = "10.0.0.0/16"

  azs             = ["us-east-1a", "us-east-1b", "us-east-1c"]
  private_subnets = ["10.0.1.0/24", "10.0.2.0/24", "10.0.3.0/24"]
  public_subnets  = ["10.0.101.0/24", "10.0.102.0/24", "10.0.103.0/24"]

  enable_nat_gateway = true
  single_nat_gateway = false
}

module "aurora" {
  source  = "terraform-aws-modules/rds-aurora/aws"
  version = "9.0.0"

  name           = "logrix-aurora-postgresql"
  engine         = "aurora-postgresql"
  engine_version = "16.1"
  serverlessv2_scaling_configuration = {
    min_capacity = 0.5
    max_capacity = 16.0
  }
```

```
  vpc_id  = module.vpc.vpc_id
  subnets = module.vpc.private_subnets
}

module "eks" {
  source  = "terraform-aws-modules/eks/aws"
  version = "19.21.0"

  cluster_name    = "logrix-eks"
  cluster_version = "1.28"

  vpc_id     = module.vpc.vpc_id
  subnet_ids = module.vpc.private_subnets

  eks_managed_node_groups = {
    workers = {
      min_size     = 2
      max_size     = 10
      desired_size = 3
      instance_types = ["c6i.xlarge"]
    }
  }
}
```